

Quantum Computing and Option Pricing

Mnacho Echenim

Grenoble INP - Ensimag, UGA

2025-2026

Outline

- 1 Quantum pricing
- 2 Storing a probability distribution in a quantum state

Some notations

- Given $(x_0, \dots, x_{n-1}) \in \mathbb{Z}_2^n$, let $x = \sum_{i=0}^{2^n-1} x_i 2^i$. We may write $|x\rangle_n$ as a shorthand for $|x_{n-1} \dots x_0\rangle = |x_{n-1}\rangle \otimes \dots \otimes |x_0\rangle$
 - Example: $|6\rangle_3 = |110\rangle$ and $|6\rangle_4 = |0110\rangle$
- We will write $|x\rangle$ instead of $|x\rangle_n$ when n is clear from the context

Some notations

- Given $(x_0, \dots, x_{n-1}) \in \mathbb{Z}_2^n$, let $x = \sum_{i=0}^{2^n-1} x_i 2^i$. We may write $|x\rangle_n$ as a shorthand for $|x_{n-1} \dots x_0\rangle = |x_{n-1}\rangle \otimes \dots \otimes |x_0\rangle$
 - Example: $|6\rangle_3 = |110\rangle$ and $|6\rangle_4 = |0110\rangle$
- We will write $|x\rangle$ instead of $|x\rangle_n$ when n is clear from the context

Qubit representation

Instead of representing a system over n qubits as

$$|\psi\rangle = \sum_{(b_0, \dots, b_{n-1}) \in \mathbb{Z}_2^n} \alpha_{b_{n-1} \dots b_0} |b_{n-1} \dots b_0\rangle,$$

we may simply write

$$|\psi\rangle = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle$$

Qubit selection

Question

Given a circuit on n qubits, how can we represent a gate on $m \leq n$ qubits in an arbitrary order, controlled by some other qubits?

Qubit selection

Question

Given a circuit on n qubits, how can we represent a gate on $m \leq n$ qubits in an arbitrary order, controlled by some other qubits?

Definition

Given a system with n qubits, an operator U acting on m qubits and a sequence $R = (i_1, \dots, i_m)$ where $1 \leq m \leq n$, $i_j \leq n - 1$ and $i_j \neq i_k$ if $j \neq k$, we denote by $U[R]$ the operator corresponding to the application of U to the sequence of qubits indexed by R

Qubit selection

Question

Given a circuit on n qubits, how can we represent a gate on $m \leq n$ qubits in an arbitrary order, controlled by some other qubits?

Definition

Given a system with n qubits, an operator U acting on m qubits and a sequence $R = (i_1, \dots, i_m)$ where $1 \leq m \leq n$, $i_j \leq n-1$ and $i_j \neq i_k$ if $j \neq k$, we denote by $U[R]$ the operator corresponding to the application of U to the sequence of qubits indexed by R

Definition

Consider a system with n qubits and two sequences $R = (i_0, \dots, i_{p-1})$ and $R' = (j_0, \dots, j_{q-1})$, where $1 \leq p \leq n$, $1 \leq q \leq n$ and $\{i_0, \dots, i_{p-1}, j_0, \dots, j_{q-1}\} \subseteq \{0, \dots, n-1\}$ is of cardinality $p+q$. We denote by $c_R[U[R']]$ the operator that applies $U[R']$, controlled by $|x_{i_0}\rangle, \dots, |x_{i_{p-1}}\rangle$

Examples

Consider a system of $n = 5$ qubits:

- If $R = (0, 2)$ and $R' = (1)$ then

$$C_R \llbracket H[R'] \rrbracket \cdot |x_4 x_3 x_2 x_1 x_0\rangle = \begin{cases} |x_4 x_3 x_2\rangle \otimes (H \cdot |x_1\rangle) \otimes |x_0\rangle & \text{if } x_0 = x_2 = 1 \\ |x_4 x_3 x_2 x_1 x_0\rangle & \text{otherwise} \end{cases}$$

Examples

Consider a system of $n = 5$ qubits:

- If $R = (0, 2)$ and $R' = (1)$ then

$$C_R \llbracket H[R'] \rrbracket \cdot |x_4 x_3 x_2 x_1 x_0\rangle = \begin{cases} |x_4 x_3 x_2\rangle \otimes (H \cdot |x_1\rangle) \otimes |x_0\rangle & \text{if } x_0 = x_2 = 1 \\ |x_4 x_3 x_2 x_1 x_0\rangle & \text{otherwise} \end{cases}$$

- If V is an operator on 3 qubits and $R'' = (4, 1, 3)$ then $C_R \llbracket V[R''] \rrbracket \cdot |x_4 x_3 x_2 x_1 x_0\rangle$ corresponds to an application of V to x_4, x_1 and x_3 (in that order) exactly when $x_0 = x_2 = 1$

Examples

Consider a system of $n = 5$ qubits:

- If $R = (0, 2)$ and $R' = (1)$ then

$$C_R \llbracket H[R'] \rrbracket \cdot |x_4 x_3 x_2 x_1 x_0\rangle = \begin{cases} |x_4 x_3 x_2\rangle \otimes (H \cdot |x_1\rangle) \otimes |x_0\rangle & \text{if } x_0 = x_2 = 1 \\ |x_4 x_3 x_2 x_1 x_0\rangle & \text{otherwise} \end{cases}$$

- If V is an operator on 3 qubits and $R'' = (4, 1, 3)$ then $C_R \llbracket V[R''] \rrbracket \cdot |x_4 x_3 x_2 x_1 x_0\rangle$ corresponds to an application of V to x_4, x_1 and x_3 (in that order) exactly when $x_0 = x_2 = 1$

Convention

We may get rid of parenthesis, and e.g., write $C_{1,2} \llbracket U[4, 3] \rrbracket$

Fair price of a derivative product

Theorem

Given an asset whose evolution is described by the stochastic differential equation

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t,$$

consider a derivative product that, at maturity T , delivers a payoff $f(S_T)$. The price at $t = 0$ of this derivative product is $p = \mathbb{E} [e^{-rT} f(S_T)]$

Fair price of a derivative product

Theorem

Given an asset whose evolution is described by the stochastic differential equation

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t,$$

consider a derivative product that, at maturity T , delivers a payoff $f(S_T)$. The price at $t = 0$ of this derivative product is $p = \mathbb{E} [e^{-rT} f(S_T)]$

Modification

In what follows we will investigate how to approximate the expectation of the undiscounted payoff $\mathbb{E} [f(S_T)]$

Fair price of a derivative product

Theorem

Given an asset whose evolution is described by the stochastic differential equation

$$\frac{dS_t}{S_t} = rdt + \sigma dW_t,$$

consider a derivative product that, at maturity T , delivers a payoff $f(S_T)$. The price at $t = 0$ of this derivative product is $p = \mathbb{E} [e^{-rT} f(S_T)]$

Modification

In what follows we will investigate how to approximate the expectation of the undiscounted payoff $\mathbb{E} [f(S_T)]$

Lemma

For $M \in \mathbb{N}$, let $p_M \stackrel{\text{def}}{=} \frac{1}{M} \sum_{i=1}^M f(S_T^{(i)})$, where each $S_T^{(i)}$ is an independent sample with the same distribution as S_T . Then $|p - p_M| = O\left(M^{-\frac{1}{2}}\right)$

Example: call option pricing

- Call option
 - ▶ Financial characteristics: strike price K , maturity T , payoff $(S_T - K)_+$
 - ▶ Pricing characteristics: spot price S_0 , risk-free rate r , volatility σ

Example: call option pricing

- Call option
 - ▶ Financial characteristics: strike price K , maturity T , payoff $(S_T - K)_+$
 - ▶ Pricing characteristics: spot price S_0 , risk-free rate r , volatility σ

- Monte-Carlo approximation of the undiscounted payoff expectation

Input: Financial characteristics K , T

Input: Pricing characteristics S_0 , r , σ

Input: Sample number M

$\text{acc} \leftarrow 0;$

for $i \leftarrow 1$ **to** M **do**

$G \leftarrow \text{sample}(\mathcal{N}(0, 1))$ // *normal distribution*;

$\widetilde{S}_T \leftarrow S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}G};$

$\text{acc} \leftarrow \text{acc} + (\widetilde{S}_T - K)_+;$

end

return $\frac{\text{acc}}{M};$

Example: call option pricing

- Call option
 - ▶ Financial characteristics: strike price K , maturity T , payoff $(S_T - K)_+$
 - ▶ Pricing characteristics: spot price S_0 , risk-free rate r , volatility σ
- Monte-Carlo approximation of the undiscounted payoff expectation

Input: Financial characteristics K, T

Input: Pricing characteristics S_0, r, σ

Input: Sample number M

```

acc ← 0;
for i ← 1 to M do
    G ← sample( $\mathcal{N}(0, 1)$ ) // normal distribution;
     $\widetilde{S}_T \leftarrow S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}G}$ ;
    acc ← acc + ( $\widetilde{S}_T - K$ )+;
end
return  $\frac{\text{acc}}{M}$ ;

```
- Error of order $O\left(M^{-\frac{1}{2}}\right)$: to divide the approximation error by a factor of 10, we must increase the sample number by a factor of 100

Example: call option pricing

- Call option
 - ▶ Financial characteristics: strike price K , maturity T , payoff $(S_T - K)_+$
 - ▶ Pricing characteristics: spot price S_0 , risk-free rate r , volatility σ

- Monte-Carlo approximation of the undiscounted payoff expectation

Input: Financial characteristics K , T

Input: Pricing characteristics S_0 , r , σ

Input: Sample number M

$\text{acc} \leftarrow 0;$

for $i \leftarrow 1$ **to** M **do**

$G \leftarrow \text{sample}(\mathcal{N}(0, 1))$ // *normal distribution*;

$\widetilde{S}_T \leftarrow S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}G};$

$\text{acc} \leftarrow \text{acc} + (\widetilde{S}_T - K)_+;$

end

return $\frac{\text{acc}}{M};$

- Error of order $O\left(M^{-\frac{1}{2}}\right)$: to divide the approximation error by a factor of 10, we must increase the sample number by a factor of 100
- Can we do better with a quantum algorithm?

The Quantum Amplitude Estimation algorithm

Input A system in the state $\sqrt{1-a}(|\psi_0\rangle_n \otimes |0\rangle) + \sqrt{a}(|\psi_1\rangle_n \otimes |1\rangle)$

Main operation Uses m qubits and applies $M = 2^m$ operations of an operator Q to the state

Output A value of the form $\tilde{a} = \sin^2\left(\frac{y\pi}{M}\right)$ with $y \in \{0, 2^m - 1\}$

The Quantum Amplitude Estimation algorithm

Input A system in the state $\sqrt{1-a}(|\Psi_0\rangle_n \otimes |0\rangle) + \sqrt{a}(|\Psi_1\rangle_n \otimes |1\rangle)$

Main operation Uses m qubits and applies $M = 2^m$ operations of an operator Q to the state

Output A value of the form $\tilde{a} = \sin^2\left(\frac{y\pi}{M}\right)$ with $y \in \{0, 2^m - 1\}$

Lemma

With probability $P \geq \frac{8}{\pi^2}$, we have $|a - \tilde{a}| = \frac{\pi}{M} + \frac{\pi^2}{M^2} = O(M^{-1})$

The Quantum Amplitude Estimation algorithm

Input A system in the state $\sqrt{1-a}(|\Psi_0\rangle_n \otimes |0\rangle) + \sqrt{a}(|\Psi_1\rangle_n \otimes |1\rangle)$

Main operation Uses m qubits and applies $M = 2^m$ operations of an operator Q to the state

Output A value of the form $\tilde{a} = \sin^2\left(\frac{y\pi}{M}\right)$ with $y \in \{0, 2^m - 1\}$

Lemma

With probability $P \geq \frac{8}{\pi^2}$, we have $|a - \tilde{a}| = \frac{\pi}{M} + \frac{\pi^2}{M^2} = O(M^{-1})$

Summary

- Given a system in the state $\sqrt{1-a}(|\Psi_0\rangle_n \otimes |0\rangle) + \sqrt{a}(|\Psi_1\rangle_n \otimes |1\rangle)$, we can estimate the probability of measuring a “1” on the last qubit in M operations with an error of order $O(M^{-1})$
- We will construct a state such that this probability is the expectation we aim to compute

Using QAE to estimate an expectation

Given a random variable X with values in $\{0, \dots, 2^n - 1\}$ and probabilities $p_i \stackrel{\text{def}}{=} \mathbb{P}(X = i)$ and a function $f : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$, QAE can be used to estimate $\mathbb{E}[f(X)]$ as follows:

- 1 Construct a state $|\psi\rangle$ of the form $\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle$

Using QAE to estimate an expectation

Given a random variable X with values in $\{0, \dots, 2^n - 1\}$ and probabilities $p_i \stackrel{\text{def}}{=} \mathbb{P}(X = i)$ and a function $f : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$, QAE can be used to estimate $\mathbb{E}[f(X)]$ as follows:

- 1 Construct a state $|\psi\rangle$ of the form $\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle$
- 2 Construct an operator F on $n+1$ qubits such that for all $i \in \{0, \dots, 2^n - 1\}$,

$$\begin{aligned} F \cdot (|i\rangle \otimes |0\rangle) &= |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\ &= \sqrt{1-f(i)} |i\rangle \otimes |0\rangle + \sqrt{f(i)} |i\rangle \otimes |1\rangle \end{aligned}$$

Using QAE to estimate an expectation

Given a random variable X with values in $\{0, \dots, 2^n - 1\}$ and probabilities $p_i \stackrel{\text{def}}{=} \mathbb{P}(X = i)$ and a function $f : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$, QAE can be used to estimate $\mathbb{E}[f(X)]$ as follows:

- 1 Construct a state $|\psi\rangle$ of the form $\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle$
- 2 Construct an operator F on $n + 1$ qubits such that for all $i \in \{0, \dots, 2^n - 1\}$,

$$\begin{aligned} F \cdot (|i\rangle \otimes |0\rangle) &= |i\rangle \otimes \left(\sqrt{1 - f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\ &= \sqrt{1 - f(i)} |i\rangle \otimes |0\rangle + \sqrt{f(i)} |i\rangle \otimes |1\rangle \end{aligned}$$

- 3 Apply F to $|\psi\rangle \otimes |0\rangle$

Using QAE to estimate an expectation

Given a random variable X with values in $\{0, \dots, 2^n - 1\}$ and probabilities $p_i \stackrel{\text{def}}{=} \mathbb{P}(X = i)$ and a function $f : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$, QAE can be used to estimate $\mathbb{E}[f(X)]$ as follows:

- 1 Construct a state $|\psi\rangle$ of the form $\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle$
- 2 Construct an operator F on $n + 1$ qubits such that for all $i \in \{0, \dots, 2^n - 1\}$,

$$\begin{aligned} F \cdot (|i\rangle \otimes |0\rangle) &= |i\rangle \otimes \left(\sqrt{1 - f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\ &= \sqrt{1 - f(i)} |i\rangle \otimes |0\rangle + \sqrt{f(i)} |i\rangle \otimes |1\rangle \end{aligned}$$

- 3 Apply F to $|\psi\rangle \otimes |0\rangle$
- 4 Use QAE to estimate the probability of measuring a “1” on the last qubit

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$F \cdot (|\psi\rangle \otimes |0\rangle) = F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right)$$

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$\begin{aligned} F \cdot (|\psi\rangle \otimes |0\rangle) &= F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right) \\ &= \sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \end{aligned}$$

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$\begin{aligned}
 F \cdot (|\psi\rangle \otimes |0\rangle) &= F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i(1-f(i))} |i\rangle \otimes |0\rangle + \sum_{i=0}^{2^n-1} \sqrt{p_i f(i)} |i\rangle \otimes |1\rangle
 \end{aligned}$$

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$\begin{aligned}
 F \cdot (|\psi\rangle \otimes |0\rangle) &= F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i(1-f(i))} |i\rangle \otimes |0\rangle + \sum_{i=0}^{2^n-1} \sqrt{p_i f(i)} |i\rangle \otimes |1\rangle
 \end{aligned}$$

$$\text{Let } a \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} p_i f(i), \quad |\Psi_0\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i(1-f(i))}}{\sqrt{1-a}} |i\rangle \quad \text{and} \quad |\Psi_1\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i f(i)}}{\sqrt{a}} |i\rangle$$

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$\begin{aligned}
 F \cdot (|\psi\rangle \otimes |0\rangle) &= F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\
 &= \sum_{i=0}^{2^n-1} \sqrt{p_i(1-f(i))} |i\rangle \otimes |0\rangle + \sum_{i=0}^{2^n-1} \sqrt{p_i f(i)} |i\rangle \otimes |1\rangle
 \end{aligned}$$

$$\text{Let } a \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} p_i f(i), \quad |\Psi_0\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i(1-f(i))}}{\sqrt{1-a}} |i\rangle \quad \text{and} \quad |\Psi_1\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i f(i)}}{\sqrt{a}} |i\rangle$$

$$\text{We have } F \cdot (|\psi\rangle \otimes |0\rangle) = \sqrt{1-a} (|\Psi_0\rangle \otimes |0\rangle) + \sqrt{a} (|\Psi_1\rangle \otimes |1\rangle)$$

Using QAE to compute an expectation (cont'd)

Proposition

QAE applied to $F \cdot (|\psi\rangle \otimes |0\rangle)$ permits to approximate $\mathbb{E}[f(X)]$

Proof. We have

$$\begin{aligned} F \cdot (|\psi\rangle \otimes |0\rangle) &= F \cdot \left(\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes |0\rangle \right) \\ &= \sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle \otimes \left(\sqrt{1-f(i)} |0\rangle + \sqrt{f(i)} |1\rangle \right) \\ &= \sum_{i=0}^{2^n-1} \sqrt{p_i(1-f(i))} |i\rangle \otimes |0\rangle + \sum_{i=0}^{2^n-1} \sqrt{p_i f(i)} |i\rangle \otimes |1\rangle \end{aligned}$$

$$\text{Let } a \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} p_i f(i), \quad |\Psi_0\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i(1-f(i))}}{\sqrt{1-a}} |i\rangle \quad \text{and} \quad |\Psi_1\rangle \stackrel{\text{def}}{=} \sum_{i=0}^{2^n-1} \frac{\sqrt{p_i f(i)}}{\sqrt{a}} |i\rangle$$

$$\text{We have } F \cdot (|\psi\rangle \otimes |0\rangle) = \sqrt{1-a} (|\Psi_0\rangle \otimes |0\rangle) + \sqrt{a} (|\Psi_1\rangle \otimes |1\rangle)$$

QAE returns an estimation of $a = \mathbb{E}[f(X)]$

QAE for call option pricing

- What follows: an encoding of the problem in a quantum setting that uses QAE to approximate the undiscounted payoff expectation with an error of order $O\left(M^{-\frac{2}{3}}\right)$
 - ▶ To divide the approximation error by a factor of 10, we must increase the sample number by a factor of ~ 30

QAE for call option pricing

- What follows: an encoding of the problem in a quantum setting that uses QAE to approximate the undiscounted payoff expectation with an error of order $O\left(M^{-\frac{2}{3}}\right)$
 - ▶ To divide the approximation error by a factor of 10, we must increase the sample number by a factor of ~ 30
 - ▶ **Can't we do better?** QAE has an error of order $O(M^{-1})$

QAE for call option pricing

- What follows: an encoding of the problem in a quantum setting that uses QAE to approximate the undiscounted payoff expectation with an error of order $O\left(M^{-\frac{2}{3}}\right)$
 - ▶ To divide the approximation error by a factor of 10, we must increase the sample number by a factor of ~ 30
 - ▶ **Can't we do better?** QAE has an error of order $O(M^{-1})$
 - ▶ The $O(M^{-1})$ error can be approached but the encoding in a quantum setting becomes much more complex

QAE for call option pricing

- What follows: an encoding of the problem in a quantum setting that uses QAE to approximate the undiscounted payoff expectation with an error of order $O\left(M^{-\frac{2}{3}}\right)$
 - ▶ To divide the approximation error by a factor of 10, we must increase the sample number by a factor of ~ 30
 - ▶ **Can't we do better?** QAE has an error of order $O(M^{-1})$
 - ▶ The $O(M^{-1})$ error can be approached but the encoding in a quantum setting becomes much more complex
- Main encoding steps:
 - ▶ Transforming the problem to phrase it in a discrete setting
 - ▶ Storing a probability distribution in a quantum setting
 - ▶ Encoding (an approximation of) a piecewise linear payoff in a quantum state
 - ▶ Encoding the call payoff in a quantum state

Problem transformation

Given a number n of qubits used for the encoding, we need to specify

- A function $f : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$
- A probability distribution $(p_i)_i$ of a discrete random variable X on $\{0, \dots, 2^n - 1\}$

Such that (an approximation of) $\mathbb{E}[(S_T - K)_+]$ can be deduced straightforwardly from $\mathbb{E}[f(X)]$

Step 1 Discretize the possible values of S_T on a regular grid of size 2^n

$$\mathbb{E}[(S_T - K)_+] \approx \sum_{i=0}^{2^n-1} \mathbb{P}(s_i \leq S_T < s_{i+1}) \cdot (s_i - K)_+$$

Step 2 Define $f_c : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$ such that, for all i ,
 $(s_i - K)_+ = \rho \cdot f_c(i)$, for some $\rho \geq 0$

Step 3 Define $p : \{0, \dots, 2^n - 1\} \rightarrow [0, 1]$ such that $\sum_{j=0}^{2^n-1} p(j) = 1$ and, for all i ,
 $p(i) \geq 0$ and $p(i) \approx \mathbb{P}(s_i \leq S_T < s_{i+1})$

We use the approximation $\mathbb{E}[(S_T - K)_+] \approx \rho \cdot \left(\sum_{i=0}^{2^n-1} p(i) \cdot f_c(i) \right)$

Discretization

- Discretization of S_T on a regular grid $(s_m, \dots, s_M)$ of size 2^n
 - ▶ Value $i \in \{0, \dots, 2^n - 1\}$ represents $s_i \stackrel{\text{def}}{=} s_m + i \frac{s_M - s_m}{2^n - 1}$

Discretization

- Discretization of S_T on a regular grid $(s_m, \dots, s_M)$ of size 2^n
 - ▶ Value $i \in \{0, \dots, 2^n - 1\}$ represents $s_i \stackrel{\text{def}}{=} s_m + i \frac{s_M - s_m}{2^n - 1}$
 - ▶ Standard practice: using $\bar{s} \stackrel{\text{def}}{=} \mathbb{E}[S_T] = S_0 e^{rT}$, set $s_m \stackrel{\text{def}}{=} \max(0, \bar{s} - 3\sigma)$ and $s_M \stackrel{\text{def}}{=} \bar{s} + 3\sigma$

Discretization

- Discretization of S_T on a regular grid $(s_m, \dots, s_M)$ of size 2^n
 - ▶ Value $i \in \{0, \dots, 2^n - 1\}$ represents $s_i \stackrel{\text{def}}{=} s_m + i \frac{s_M - s_m}{2^n - 1}$
 - ▶ Standard practice: using $\bar{s} \stackrel{\text{def}}{=} \mathbb{E}[S_T] = S_0 e^{rT}$, set $s_m \stackrel{\text{def}}{=} \max(0, \bar{s} - 3\sigma)$ and $s_M \stackrel{\text{def}}{=} \bar{s} + 3\sigma$
 - ▶ We assume that K belongs to this grid: $K = s_L$ for $L \in \{0, \dots, 2^n - 1\}$; in particular $s_m \leq K \leq s_M$ and the payoff is of the form $(s_i - s_L)_+$ for some $i \in \{0, \dots, 2^n - 1\}$

Discretization

- Discretization of S_T on a regular grid $(s_m, \dots, s_M)$ of size 2^n
 - Value $i \in \{0, \dots, 2^n - 1\}$ represents $s_i \stackrel{\text{def}}{=} s_m + i \frac{s_M - s_m}{2^n - 1}$
 - Standard practice: using $\bar{s} \stackrel{\text{def}}{=} \mathbb{E}[S_T] = S_0 e^{rT}$, set $s_m \stackrel{\text{def}}{=} \max(0, \bar{s} - 3\sigma)$ and $s_M \stackrel{\text{def}}{=} \bar{s} + 3\sigma$
 - We assume that K belongs to this grid: $K = s_L$ for $L \in \{0, \dots, 2^n - 1\}$; in particular $s_m \leq K \leq s_M$ and the payoff is of the form $(s_i - s_L)_+$ for some $i \in \{0, \dots, 2^n - 1\}$
- Affine transformation of the payoff
 - Using the fact that $(s_M - K)_+ = s_M - K = s_M - s_L$, define the function

$$f_c : \begin{array}{ll} \{0, \dots, 2^n - 1\} & \rightarrow [0, 1] \\ i & \mapsto \frac{(s_i - K)_+}{s_M - K} = \frac{(s_i - s_L)_+}{s_M - s_L} \end{array}$$

Discretization

- Discretization of S_T on a regular grid $(s_m, \dots, s_M)$ of size 2^n
 - Value $i \in \{0, \dots, 2^n - 1\}$ represents $s_i \stackrel{\text{def}}{=} s_m + i \frac{s_M - s_m}{2^n - 1}$
 - Standard practice: using $\bar{s} \stackrel{\text{def}}{=} \mathbb{E}[S_T] = S_0 e^{rT}$, set $s_m \stackrel{\text{def}}{=} \max(0, \bar{s} - 3\sigma)$ and $s_M \stackrel{\text{def}}{=} \bar{s} + 3\sigma$
 - We assume that K belongs to this grid: $K = s_L$ for $L \in \{0, \dots, 2^n - 1\}$; in particular $s_m \leq K \leq s_M$ and the payoff is of the form $(s_i - s_L)_+$ for some $i \in \{0, \dots, 2^n - 1\}$
- Affine transformation of the payoff
 - Using the fact that $(s_M - K)_+ = s_M - K = s_M - s_L$, define the function

$$f_c : \begin{array}{ll} \{0, \dots, 2^n - 1\} & \rightarrow [0, 1] \\ i & \mapsto \frac{(s_i - K)_+}{s_M - K} = \frac{(s_i - s_L)_+}{s_M - s_L} \end{array}$$
 - Hence, $(s_i - s_L)_+ = (s_M - s_L)f_c(i)$

Probability distribution

- Define $s_{2^n} \stackrel{\text{def}}{=} s_M + \frac{s_M - s_m}{2^n - 1}$

Probability distribution

- Define $s_{2^n} \stackrel{\text{def}}{=} s_M + \frac{s_M - s_m}{2^n - 1}$
- For $i \in \{0, \dots, 2^n - 1\}$, set

$$p_i \stackrel{\text{def}}{=} \frac{\mathbb{P}(s_i \leq S_T < s_{i+1})}{\mathbb{P}(s_m \leq S_T < s_{2^n})}$$

Probability distribution

- Define $s_{2^n} \stackrel{\text{def}}{=} s_M + \frac{s_M - s_m}{2^n - 1}$
- For $i \in \{0, \dots, 2^n - 1\}$, set

$$p_i \stackrel{\text{def}}{=} \frac{\mathbb{P}(s_i \leq S_T < s_{i+1})}{\mathbb{P}(s_m \leq S_T < s_{2^n})}$$

- Note that $\mathbb{P}(s_m \leq S_T < s_{2^n}) \approx 1$, hence $p_i \approx \mathbb{P}(s_i \leq S_T < s_{i+1})$

Probability distribution

- Define $s_{2^n} \stackrel{\text{def}}{=} s_M + \frac{s_M - s_m}{2^n - 1}$
- For $i \in \{0, \dots, 2^n - 1\}$, set

$$p_i \stackrel{\text{def}}{=} \frac{\mathbb{P}(s_i \leq S_T < s_{i+1})}{\mathbb{P}(s_m \leq S_T < s_{2^n})}$$

- Note that $\mathbb{P}(s_m \leq S_T < s_{2^n}) \approx 1$, hence $p_i \approx \mathbb{P}(s_i \leq S_T < s_{i+1})$

Proposition

If X is a random variable such that for all $i \in \{0, \dots, 2^n - 1\}$ we have $\mathbb{P}(X = i) = p_i$, then we have the following:

$$(s_M - s_L) \cdot \sum_{i=0}^{2^n - 1} p_i f_c(i) = (s_M - s_L) \cdot \mathbb{E}[f_c(X)] \approx \mathbb{E}[(S_T - K)_+]$$

Probability distribution

- Define $s_{2^n} \stackrel{\text{def}}{=} s_M + \frac{s_M - s_m}{2^n - 1}$
- For $i \in \{0, \dots, 2^n - 1\}$, set

$$p_i \stackrel{\text{def}}{=} \frac{\mathbb{P}(s_i \leq S_T < s_{i+1})}{\mathbb{P}(s_m \leq S_T < s_{2^n})}$$

- Note that $\mathbb{P}(s_m \leq S_T < s_{2^n}) \approx 1$, hence $p_i \approx \mathbb{P}(s_i \leq S_T < s_{i+1})$

Proposition

If X is a random variable such that for all $i \in \{0, \dots, 2^n - 1\}$ we have $\mathbb{P}(X = i) = p_i$, then we have the following:

$$(s_M - s_L) \cdot \sum_{i=0}^{2^n-1} p_i f_c(i) = (s_M - s_L) \cdot \mathbb{E}[f_c(X)] \approx \mathbb{E}[(S_T - K)_+]$$

There remains to store p and f_c in quantum states to be able to use the QPE algorithm

Outline

- 1 Quantum pricing
- 2 Storing a probability distribution in a quantum state

A sequence of probability distributions

Task at hand

Problem statement

Input A set $\{p_i \mid i = 0, \dots, 2^n - 1\}$

Goal Create a state $\sum_{i=0}^{2^n-1} \sqrt{p_i} |i\rangle$

Equivalent problem statement

Input A set $\{p_w \mid w \in \mathbb{Z}_2^n\}$

Goal Create a state $\sum_{w \in \mathbb{Z}_2^n} \sqrt{p_w} |w\rangle$

Case where $n = 1$

- **Input:** Positive numbers p_0 and p_1 , where $p_0 + p_1 = 1$
- **Goal:** Create the state $\sqrt{p_0}|0\rangle + \sqrt{p_1}|1\rangle$

Case where $n = 1$

- **Input:** Positive numbers p_0 and p_1 , where $p_0 + p_1 = 1$
- **Goal:** Create the state $\sqrt{p_0}|0\rangle + \sqrt{p_1}|1\rangle$
- **Method:** By using $\mathcal{R}_\alpha^y$, the rotation gate around the y -axis
 - ▶ The matrix representation of $\mathcal{R}_\alpha^y$ is

$$\mathcal{R}_\alpha^y = \begin{pmatrix} \cos\left(\frac{\alpha}{2}\right) & -\sin\left(\frac{\alpha}{2}\right) \\ \sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{pmatrix}$$

Case where $n = 1$

- **Input:** Positive numbers p_0 and p_1 , where $p_0 + p_1 = 1$
- **Goal:** Create the state $\sqrt{p_0}|0\rangle + \sqrt{p_1}|1\rangle$
- **Method:** By using $\mathcal{R}_\alpha^y$, the rotation gate around the y -axis
 - ▶ The matrix representation of $\mathcal{R}_\alpha^y$ is

$$\mathcal{R}_\alpha^y = \begin{pmatrix} \cos\left(\frac{\alpha}{2}\right) & -\sin\left(\frac{\alpha}{2}\right) \\ \sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{pmatrix}$$

- ▶ Let $\theta \stackrel{\text{def}}{=} 2 \arccos(\sqrt{p_0})$, then we have

$$\begin{aligned} \mathcal{R}_\theta^y \cdot |0\rangle &= \cos\left(\frac{\theta}{2}\right) |0\rangle + \sin\left(\frac{\theta}{2}\right) |1\rangle \\ &= \sqrt{p_0}|0\rangle + \sqrt{1-p_0}|1\rangle \\ &= \sqrt{p_0}|0\rangle + \sqrt{p_1}|1\rangle \end{aligned}$$

Case where $n = 2$

- **Input:** Positive numbers p_{00} , p_{01} , p_{10} and p_{11} , where $\sum_{w \in \mathbb{Z}_2^2} p_w = 1$
- **Goal:** Create the state $\sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$

Case where $n = 2$

- **Input:** Positive numbers p_{00} , p_{01} , p_{10} and p_{11} , where $\sum_{w \in \mathbb{Z}_2^2} p_w = 1$
- **Goal:** Create the state $\sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$
- **Method:** let $q_0 \stackrel{\text{def}}{=} p_{00} + p_{01}$ and $q_1 \stackrel{\text{def}}{=} p_{10} + p_{11}$
 - ▶ We know how to construct the state $\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle$ using a rotation gate with angle $\theta \stackrel{\text{def}}{=} 2 \arccos(\sqrt{q_0})$

Case where $n = 2$

- **Input:** Positive numbers p_{00} , p_{01} , p_{10} and p_{11} , where $\sum_{w \in \mathbb{Z}_2^2} p_w = 1$
- **Goal:** Create the state $\sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$
- **Method:** let $q_0 \stackrel{\text{def}}{=} p_{00} + p_{01}$ and $q_1 \stackrel{\text{def}}{=} p_{10} + p_{11}$
 - ▶ We know how to construct the state $\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle$ using a rotation gate with angle $\theta \stackrel{\text{def}}{=} 2 \arccos(\sqrt{q_0})$
 - ▶ We define the angles $\theta_0 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{00}}{q_0}}\right)$ and $\theta_1 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{10}}{q_1}}\right)$, we have

$$\begin{aligned}
 (\mathbb{I} \otimes \mathcal{R}_{\theta_0}^y) \cdot (\sqrt{q_0}|0\rangle \otimes |0\rangle) &= \sqrt{q_0} \left(\cos\left(\frac{\theta_0}{2}\right) |00\rangle + \sin\left(\frac{\theta_0}{2}\right) |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{1 - \frac{p_{00}}{q_0}} |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{\frac{p_{01}}{q_0}} |01\rangle \right) \\
 &= \sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle
 \end{aligned}$$

Case where $n = 2$

- **Input:** Positive numbers p_{00} , p_{01} , p_{10} and p_{11} , where $\sum_{w \in \mathbb{Z}_2^2} p_w = 1$
- **Goal:** Create the state $\sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$
- **Method:** let $q_0 \stackrel{\text{def}}{=} p_{00} + p_{01}$ and $q_1 \stackrel{\text{def}}{=} p_{10} + p_{11}$
 - ▶ We know how to construct the state $\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle$ using a rotation gate with angle $\theta \stackrel{\text{def}}{=} 2 \arccos(\sqrt{q_0})$
 - ▶ We define the angles $\theta_0 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{00}}{q_0}}\right)$ and $\theta_1 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{10}}{q_1}}\right)$, we have

$$\begin{aligned}
 (\mathbf{I} \otimes \mathcal{R}_{\theta_0}^y) \cdot (\sqrt{q_0}|0\rangle \otimes |0\rangle) &= \sqrt{q_0} \left(\cos\left(\frac{\theta_0}{2}\right) |00\rangle + \sin\left(\frac{\theta_0}{2}\right) |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{1 - \frac{p_{00}}{q_0}} |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{\frac{p_{01}}{q_0}} |01\rangle \right) \\
 &= \sqrt{p_{00}} |00\rangle + \sqrt{p_{01}} |01\rangle \\
 (\mathbf{I} \otimes \mathcal{R}_{\theta_1}^y) \cdot (\sqrt{q_1}|1\rangle \otimes |0\rangle) &= \sqrt{p_{10}} |10\rangle + \sqrt{p_{11}} |11\rangle
 \end{aligned}$$

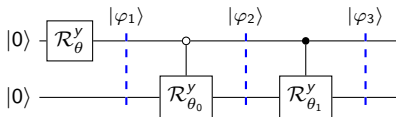
Case where $n = 2$

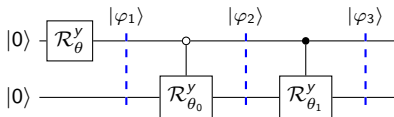
- **Input:** Positive numbers p_{00} , p_{01} , p_{10} and p_{11} , where $\sum_{w \in \mathbb{Z}_2^2} p_w = 1$
- **Goal:** Create the state $\sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$
- **Method:** let $q_0 \stackrel{\text{def}}{=} p_{00} + p_{01}$ and $q_1 \stackrel{\text{def}}{=} p_{10} + p_{11}$
 - ▶ We know how to construct the state $\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle$ using a rotation gate with angle $\theta \stackrel{\text{def}}{=} 2 \arccos(\sqrt{q_0})$
 - ▶ We define the angles $\theta_0 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{00}}{q_0}}\right)$ and $\theta_1 \stackrel{\text{def}}{=} 2 \arccos\left(\sqrt{\frac{p_{10}}{q_1}}\right)$, we have

$$\begin{aligned}
 (\mathbf{I} \otimes \mathcal{R}_{\theta_0}^y) \cdot (\sqrt{q_0}|0\rangle \otimes |0\rangle) &= \sqrt{q_0} \left(\cos\left(\frac{\theta_0}{2}\right) |00\rangle + \sin\left(\frac{\theta_0}{2}\right) |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{1 - \frac{p_{00}}{q_0}} |01\rangle \right) \\
 &= \sqrt{q_0} \left(\sqrt{\frac{p_{00}}{q_0}} |00\rangle + \sqrt{\frac{p_{01}}{q_0}} |01\rangle \right) \\
 &= \sqrt{p_{00}} |00\rangle + \sqrt{p_{01}} |01\rangle \\
 (\mathbf{I} \otimes \mathcal{R}_{\theta_1}^y) \cdot (\sqrt{q_1}|1\rangle \otimes |0\rangle) &= \sqrt{p_{10}} |10\rangle + \sqrt{p_{11}} |11\rangle
 \end{aligned}$$

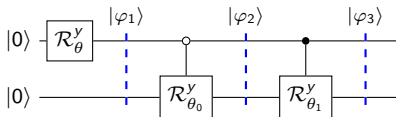
- ▶ **Next step:** combine the three rotations to construct the required state using controlled operations

Case where $n = 2$ (cont'd)



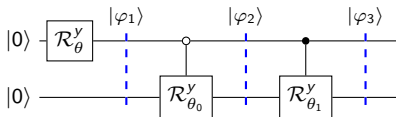
Case where $n = 2$ (cont'd)

$$\varphi_1 = (\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle) \otimes |0\rangle = \sqrt{q_0}|00\rangle + \sqrt{q_1}|10\rangle$$

Case where $n = 2$ (cont'd)

$$\varphi_1 = (\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle) \otimes |0\rangle = \sqrt{q_0}|00\rangle + \sqrt{q_1}|10\rangle$$

$$\varphi_2 = \sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{q_1}|10\rangle$$

Case where $n = 2$ (cont'd)

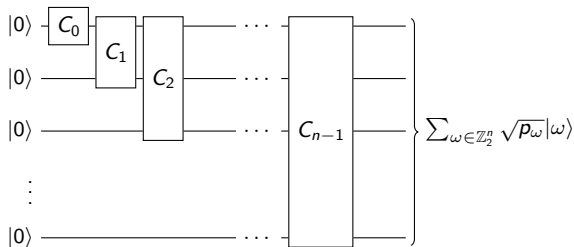
$$\varphi_1 = (\sqrt{q_0}|0\rangle + \sqrt{q_1}|1\rangle) \otimes |0\rangle = \sqrt{q_0}|00\rangle + \sqrt{q_1}|10\rangle$$

$$\varphi_2 = \sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{q_1}|10\rangle$$

$$\varphi_3 = \sqrt{p_{00}}|00\rangle + \sqrt{p_{01}}|01\rangle + \sqrt{p_{10}}|10\rangle + \sqrt{p_{11}}|11\rangle$$

General principle

- Iteratively apply controlled y -rotations



- For each $0 \leq k < n$, $C_k \cdot |0^{k+1}\rangle$ will be of the form $\sum_{w \in \mathbb{Z}_2^{k+1}} \sqrt{q_w} |w\rangle$, for some intermediate probability distribution $(q_w)_{w \in \mathbb{Z}_2^{k+1}}$

Intermediate probability distributions and rotation angles

Definition

For all $w' \in \mathbb{Z}_2^n$ let $q_{w'}^{[n]} \stackrel{\text{def}}{=} p_{w'}$, and for all $0 < m < n$ and $w \in \mathbb{Z}_2^m$, let

$$q_w^{[m]} \stackrel{\text{def}}{=} q_{w \cdot 0}^{[m+1]} + q_{w \cdot 1}^{[m+1]}, \quad \theta_w^{[m]} \stackrel{\text{def}}{=} 2 \arccos \left(\sqrt{\frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} \right) \quad \text{and} \quad \theta_\varepsilon^{[0]} \stackrel{\text{def}}{=} 2 \arccos \left(\sqrt{q_0^{[1]}} \right)$$

Intermediate probability distributions and rotation angles

Definition

For all $w' \in \mathbb{Z}_2^n$ let $q_{w'}^{[n]} \stackrel{\text{def}}{=} p_{w'}$, and for all $0 < m < n$ and $w \in \mathbb{Z}_2^m$, let

$$q_w^{[m]} \stackrel{\text{def}}{=} q_{w \cdot 0}^{[m+1]} + q_{w \cdot 1}^{[m+1]}, \quad \theta_w^{[m]} \stackrel{\text{def}}{=} 2 \arccos \left(\sqrt{\frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} \right) \quad \text{and} \quad \theta_\varepsilon^{[0]} \stackrel{\text{def}}{=} 2 \arccos \left(\sqrt{q_0^{[1]}} \right)$$

Proposition

We have the following results, for all $0 < m < n$ and $w \in \mathbb{Z}_2^m$

$$\mathcal{R}_{\theta_\varepsilon^{[0]}}^y \cdot |0\rangle = \sqrt{q_0^{[1]}} |0\rangle + \sqrt{q_1^{[1]}} |1\rangle$$

$$\mathcal{R}_{\theta_w^{[m]}}^y \cdot |0\rangle = \sqrt{\frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} |0\rangle + \sqrt{\frac{q_{w \cdot 1}^{[m+1]}}{q_w^{[m]}}} |1\rangle$$

$$(\mathbf{I}^{\otimes m} \otimes \mathcal{R}_{\theta_w^{[m]}}^y) \cdot \left(\sqrt{q_w^{[m]}} |w\rangle \otimes |0\rangle \right) = \sqrt{q_{w \cdot 0}^{[m+1]}} |w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}} |w \cdot 1\rangle$$

Rotations with angle $\theta_w^{[m]}$

Proof. The proof is straightforward for the y-rotation with angle $\theta_\varepsilon^{[0]}$. Assume $0 < m < n$ and $w \in \mathbb{Z}_2^m$, then

$$\begin{aligned}
 \mathcal{R}_{\theta_w^{[m]}}^y \cdot |0\rangle &= \cos\left(\frac{\theta_w^{[m]}}{2}\right) |0\rangle + \sin\left(\frac{\theta_w^{[m]}}{2}\right) |1\rangle \\
 &= \sqrt{\frac{q_w^{[m+1] \cdot 0}}{q_w^{[m]}}} |0\rangle + \sqrt{1 - \frac{q_w^{[m+1] \cdot 0}}{q_w^{[m]}}} |1\rangle \\
 &= \sqrt{\frac{q_w^{[m+1] \cdot 0}}{q_w^{[m]}}} |0\rangle + \sqrt{\frac{q_w^{[m+1] \cdot 1}}{q_w^{[m]}}} |1\rangle
 \end{aligned}$$

Rotations with angle $\theta_w^{[m]}$

Proof. The proof is straightforward for the y-rotation with angle $\theta_\varepsilon^{[0]}$. Assume $0 < m < n$ and $w \in \mathbb{Z}_2^m$, then

$$\begin{aligned}
 \mathcal{R}_{\theta_w^{[m]}}^y \cdot |0\rangle &= \cos\left(\frac{\theta_w^{[m]}}{2}\right) |0\rangle + \sin\left(\frac{\theta_w^{[m]}}{2}\right) |1\rangle \\
 &= \sqrt{\frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} |0\rangle + \sqrt{1 - \frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} |1\rangle \\
 &= \sqrt{\frac{q_{w \cdot 0}^{[m+1]}}{q_w^{[m]}}} |0\rangle + \sqrt{\frac{q_{w \cdot 1}^{[m+1]}}{q_w^{[m]}}} |1\rangle \\
 (\mathbf{I}^{\otimes m} \otimes \mathcal{R}_{\theta_w^{[m]}}^y) \cdot \left(\sqrt{q_w^{[m]}} |w\rangle \otimes |0\rangle \right) &= \sqrt{q_w^{[m]}} |w\rangle \otimes \left(\mathcal{R}_{\theta_w^{[m]}}^y \cdot |0\rangle \right) \\
 &= \sqrt{q_{w \cdot 0}^{[m+1]}} (|w\rangle \otimes |0\rangle) + \sqrt{q_{w \cdot 1}^{[m+1]}} (|w\rangle \otimes |1\rangle) \\
 &= \sqrt{q_{w \cdot 0}^{[m+1]}} |w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}} |w \cdot 1\rangle
 \end{aligned}$$

Recap

- The previous proposition shows that an appropriate y -rotation applied to $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$ generates $\sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle$

Recap

- The previous proposition shows that an appropriate y -rotation applied to $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$ generates $\sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle$
- We want to define an operation that, applied to $\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$, generates

$$\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle = \sum_{w' \in \mathbb{Z}_2^{m+1}} \sqrt{q_{w'}^{[m+1]}}|w'\rangle$$

Recap

- The previous proposition shows that an appropriate y -rotation applied to $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$ generates $\sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle$
- We want to define an operation that, applied to $\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$, generates

$$\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle = \sum_{w' \in \mathbb{Z}_2^{m+1}} \sqrt{q_{w'}^{[m+1]}}|w'\rangle$$

- Question:** how do we apply the correct y -rotation to each $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$, where $w \in \mathbb{Z}_2^m$?

Recap

- The previous proposition shows that an appropriate y -rotation applied to $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$ generates $\sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle$
- We want to define an operation that, applied to $\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$, generates

$$\sum_{w \in \mathbb{Z}_2^m} \sqrt{q_{w \cdot 0}^{[m+1]}}|w \cdot 0\rangle + \sqrt{q_{w \cdot 1}^{[m+1]}}|w \cdot 1\rangle = \sum_{w' \in \mathbb{Z}_2^{m+1}} \sqrt{q_{w'}^{[m+1]}}|w'\rangle$$

- **Question:** how do we apply the correct y -rotation to each $\sqrt{q_w^{[m]}}|w\rangle \otimes |0\rangle$, where $w \in \mathbb{Z}_2^m$?
- **Answer:** by making use of *controlled* y -rotations

Applying an operation to a specific $|w\rangle$

We extend the controlled operation in a special case:

- $R = (1, \dots, m)$
- U is a single-qubit gate
- Given a word $w \in \mathbb{Z}_2^m$, we consider $C_R^w [[U[0]]]$, the operation on $|x_m \cdots x_0\rangle$ that applies U to qubit $|x_0\rangle$ if and only if $w = x_m \cdots x_1$

Applying an operation to a specific $|w\rangle$

We extend the controlled operation in a special case:

- $R = (1, \dots, m)$
- U is a single-qubit gate
- Given a word $w \in \mathbb{Z}_2^m$, we consider $C_R^w[U[0]]$, the operation on $|x_m \cdots x_0\rangle$ that applies U to qubit $|x_0\rangle$ if and only if $w = x_m \cdots x_1$

Proposition

We have the following result:

$$C_R^w[U[0]] \cdot \left(\sum_{y \in \mathbb{Z}_2^m} (\alpha_y |y\rangle \otimes |0\rangle) \right) = \alpha_w |w\rangle \otimes (U \cdot |0\rangle) + \sum_{\substack{y \in \mathbb{Z}_2^m \\ y \neq w}} (\alpha_y |y\rangle \otimes |0\rangle)$$

Applying an operation to a specific $|w\rangle$ (cont'd)

Lemma

We associate with each $w \in \mathbb{Z}_2^m$ a single-qubit gate U_w . Then we have the following result:

$$\left(\prod_{w \in \mathbb{Z}_2^m} c_R^w [[U_w[0]]] \right) \cdot \left(\sum_{y \in \mathbb{Z}_2^m} (\alpha_y |y\rangle \otimes |0\rangle) \right) = \sum_{w \in \mathbb{Z}_2^m} (\alpha_w |w\rangle \otimes (U_w \cdot |0\rangle))$$

Applying an operation to a specific $|w\rangle$ (cont'd)

Lemma

We associate with each $w \in \mathbb{Z}_2^m$ a single-qubit gate U_w . Then we have the following result:

$$\left(\prod_{w \in \mathbb{Z}_2^m} c_R^w [U_w[0]] \right) \cdot \left(\sum_{y \in \mathbb{Z}_2^m} (\alpha_y |y\rangle \otimes |0\rangle) \right) = \sum_{w \in \mathbb{Z}_2^m} (\alpha_w |w\rangle \otimes (U_w \cdot |0\rangle))$$

Corollary

Let $C_m \stackrel{\text{def}}{=} \prod_{y \in \mathbb{Z}_2^m} c_R^y \left[\mathcal{R}_{\theta_y^{[m]}}^y[0] \right]$, then

$$C_m \cdot \left(\sum_{y \in \mathbb{Z}_2^m} \left(\sqrt{q_y^{[m]}} |y\rangle \otimes |0\rangle \right) \right) = \sum_{w \in \mathbb{Z}_2^{m+1}} \sqrt{q_w^{[m+1]}} |w\rangle$$

Storing a probability distribution

Theorem

Let $C_0 \stackrel{\text{def}}{=} \mathcal{R}_{\theta_{\varepsilon}^{[0]}}^y$ and $D_n = \prod_{m=n-1}^0 (C_m \otimes \mathbf{I}^{\otimes n-m-1})$, we have

$$D_n \cdot |0^n\rangle = \sum_{w \in \mathbb{Z}_2^n} \sqrt{p_w} |w\rangle$$

Storing a probability distribution

Theorem

Let $C_0 \stackrel{\text{def}}{=} \mathcal{R}_{\theta_\varepsilon^{[0]}}^y$ and $D_n = \prod_{m=n-1}^0 (C_m \otimes \mathbf{I}^{\otimes n-m-1})$, we have

$$D_n \cdot |0^n\rangle = \sum_{w \in \mathbb{Z}_2^n} \sqrt{p_w} |w\rangle$$

Circuit for $n = 3$

